Software Requirements Specification

for

Real-Time Market Data Forecasting

Version 1.0 approved

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Revision History

Version	Date of Release	Pages Affected	Reasons for Change	Signature
1.0	18-12-2024	All	First Release	

Table of Contents

1.0 Title of the project

Real-Time Market Data Forecasting

2.0 Introduction

2.1 Purpose

This document defines the software requirements for the Real-Time Market Data Forecasting system, which processes live market data to provide real-time forecasts. The SRS includes functional, nonfunctional, and interface requirements to ensure comprehensive coverage.

2.2 Document Conventions

- Key terms are italicized.
- Mandatory requirements are prefixed with REQ.
- Placeholder terms are marked with TBD.

2.3 Intended Audience and Reading Suggestions

This document is intended for developers, project managers, stakeholders, testers, and end-users. Begin with Section 1 for context, then proceed to detailed requirements in Sections 3 and 4.

2.4 Project/Product Scope

The system will analyze live market data to forecast trends in real-time, supporting strategic decision-making and investment planning. It will provide dashboards, and predictive analytics.

2.5 References

- IEEE Std 830-1998: Recommended Practice for Software Requirements Specifications.
- [Additional references TBD]

3.0 Overall Description

3.1 Product Perspective

The system integrates with existing market data platforms via APIs, analyzing incoming data streams using machine learning models. It is a standalone application designed for high performance.

3.2 Product Functions

- Data ingestion and real-time processing.
- Predictive analysis using machine learning models.
- User interface for data visualization and alerts.

3.3 User Classes and Characteristics

- Traders: Require fast, actionable insights.
- Analysts: Use predictive models for strategy.
- Managers: Review aggregated reports.

3.4 Operating Environment

- Cloud-based infrastructure supporting APIs and analytics.
- Compatible with modern web browsers (Chrome, Edge).

3.5 Design and Implementation Constraints

- Compliance with financial regulations (e.g., GDPR).
- Real-time processing within 30s latency.

3.6 User Documentation

User manuals, FAQs, and onboarding guides.

3.7 Assumptions and Dependencies

- Availability of high-quality, real-time data streams.
- Dependency on external APIs for data sources.

4.0 External Interface Requirements

4.1 User Interfaces

Web-based dashboards with interactive charts.

4.2 Hardware Interfaces

Cloud servers for processing and storage.

4.3 Software Interfaces

Integration with third-party APIs for market data.

4.4 Communications Interfaces

HTTPS protocols for secure data transmission.

5.0 System Features

5.1 Feature 1: Real-Time Data Ingestion

Description and Priority: High-priority feature ensuring data is ingested and processed without delay.

Stimulus/Response Sequences:
- Input: Market data stream.
- Output: Processed data available within 30s.

Functional Requirements:
- REQ-1: System must handle 10,000 transactions/minute.
- REQ-2: Ensure redundancy for data availability.

5.2 Feature 2: Predictive Analytics

Description and Priority: Provides actionable insights using ML models.

Stimulus/Response Sequences:
- Input: Historical and real-time data.
- Output: Forecast reports and alerts.

Functional Requirements:
- REQ-3: Accuracy of predictions >80%.
- REQ-4: Models must update daily.

6.0 Nonfunctional Requirements

6.1 Performance Requirements

System latency must not exceed 30s for data processing.

6.2 Safety Requirements

Ensure data integrity during transmission.

6.3 Security Requirements

- Implement multi-factor authentication.
- Encrypt data at rest and in transit.

6.4 Software Quality Attributes

- Scalability to handle increased data loads.
- Maintainability for seamless updates.

6.5 Business Rules

Alerts must adhere to user-defined thresholds.

7.0 Acceptance Criteria

- System must pass all unit and integration tests.
- Latency metrics must meet specified thresholds.

8.0 Deliverables

- Fully deployed system.
- Documentation (user and technical).

9.0 Appendices

Appendix A: Glossary

Market Data: Data about stock prices, trends, etc.
ML Models: Machine Learning Models.

Appendix B: Analysis Models

[TBD]

Appendix C: To Be Determined List

Final performance benchmarks: TBD.